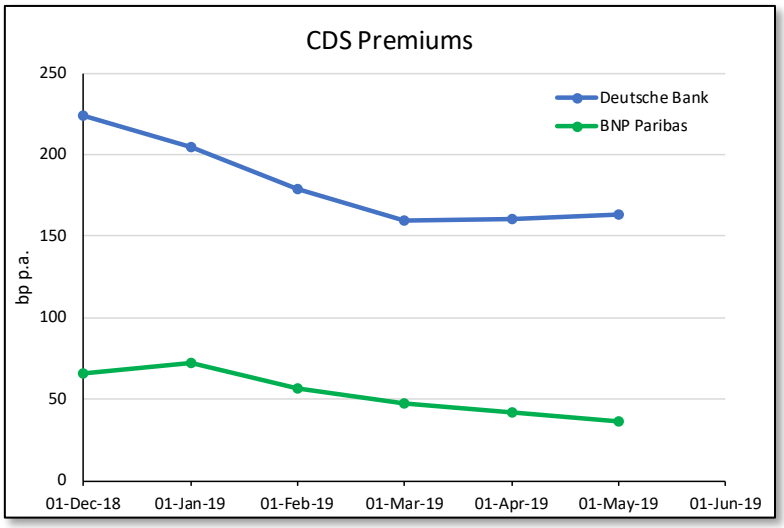
Using Credit Default Swaps

The Assignment

It is Wednesday 1st May 2019. A hedge fund investor has been monitoring CDS prices over the last few months for two large European banks – Deutsche Bank (DB) and BNP Paribas (BNP) – as pictured below.



The banking sector in Europe has been steadily improving, especially for Deutsche Bank, which has faced a number of challenges in recent years, but may have turned the corner, as evidenced by finally generating a profit of €340m in 2018 after several years of losses.

The improvement in DB's fortunes has been reflected in the pricing of CDS contracts on their name, which have tightened from 224bp p.a. in December 2018 to around 160 bp p.a. in March, where it remained for several months.

While the CDS market appears to believe that further improvements in DB's prospects are on hold, our investor believes that there is scope for a significant further tightening in their CDS premium quotations over the next month or two. However, the investor is not sure about the prospects for the European banking sector as a whole. Rather than taking an outright view on DB, our investor wants to exploit a *relative* view – that DB will strengthen against BNP. It has approached you for a trade idea to exploit their view…

On 1st May 2019, the two CDS maturing June 2024 are being quoted at:

DB 5yr CDS:	163 – 165 bp p.a. Act/360	(SDV01: €4,734.29)
BNP 5yr CDS:	36 – 38 bp p.a. Act/360	(SDV01: €5,277.69)

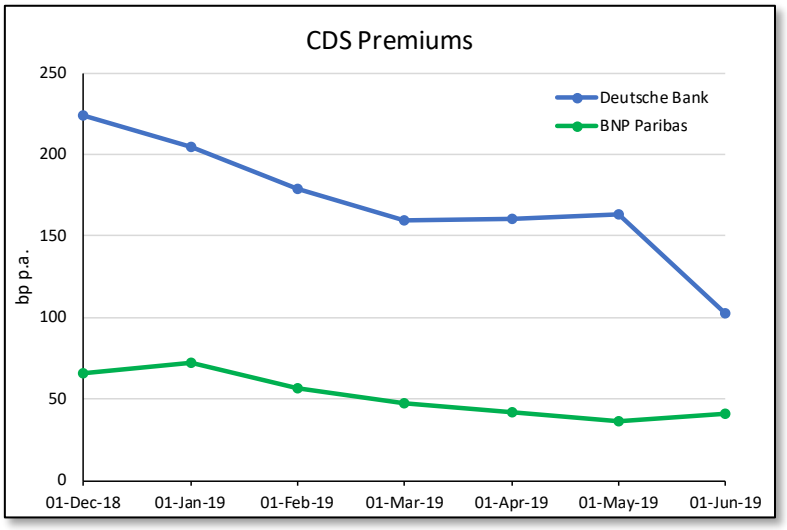
The SDV01 figures given above measure the sensitivity of each CDS position to a 1bp change in CDS premiums, assuming a notional of €10m.

Remember that the hedge fund will be paying the bid-offer spread on any transactions that it undertakes. Finally, both names trade with a 100bp standard premium.

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REQUIRED

1. The hedge fund has told you that they want to execute a €10m trade in each CDS to take advantage of their view that DB credit will improve relative to that of BNP. Which CDS should they buy, and which one should they sell?
2. Using the SDV01s on the previous page, what will be the approximate mark-to-market profit on the €10m trades if the spread between DB and BNP CDSs tightens by 1bp, either: (a) If DB premiums fall by 1bp, *or* (b) if BNP premiums rise by 1bp? Ignore bid-offer spreads in your answer.
3. If the premiums for all banking sector CDS contracts were to decline by 10bp, what would be the impact?
4. By early June 2019, CDS premiums for Deutsche Bank has tightened considerably, even though that for BNP Paribas has widened slightly.



Specifically, on Wednesday 5th June 2019, CDS premiums are now:

DB 5yr CDS: 103 – 105 bp p.a. Act/360
BNP 5yr CDS: 41 – 43 bp p.a. Act/360

Given the data in the table below for the ***protection buyer's*** up-front cashflows on €10m trades, what are the exact profits on the trading strategy?

	DB			BNP		
1 May 2019	Off-market	(163 bp):	−306,836	Off-market	(36 bp):	329,111
	Off-market	(165 bp):	−316,309	Off-market	(38 bp):	318,551
	Accrued	(43 days):	11,944	Accrued	(43 days):	11,944
5 Jun 2019	Off-market	(103 bp):	−14,770	Off-market	(41 bp):	298,186
	Off-market	(105 bp):	−24,596	Off-market	(43 bp):	287,834
	Accrued	(78 days):	21,667	Accrued	(78 days):	21,667

(A negative figure in the table means that the protection buyer pays, while a positive figure means that the protection buyer receives the cash flow. For the protection seller's cash flows, just reverse the sign of the numbers in the table.)

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